

ANXIETY DISORDER

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DEFINITION OF ANXIETY

Anxiety is a tense emotional state associated with a feeling of impending danger, often accompanied by somatic symptoms. It is used to describe the mental and physical response to a feared situation - **Flight or Fight**.

Anxiety is normal and serves as a built-in warning device. Moderate levels of anxiety can enhance performance. Even high levels of anxiety are normal if consistent with the demands of the situation.

SYMPTOMS OF ANXIETY

1. Psychological

- Irritability
- Poor concentration and memory
- Restlessness
- Worrying thoughts
- Sexual Dysfunction
- Insomnia / Nightmares

2. Physical

- Bowel disturbance
- Tremor
- Indigestion
- Dizziness
- Chest discomfort
- Headache
- Difficulty inhaling
- Muscle ache
- Palpitations

5 BASIC QUESTIONS FOR PATIENT ASSESSMENT

These are the questions that a doctor should ask himself or herself when a patient presents with anxiety symptoms.

- Is what my patient experiencing **P**athological?
- What is the **P**attern of the symptoms described?
- What are the present stressors and **P**roblems faced by him/her?
- What can I do for him / her **P**ractically in a busy practice?
- Is **P**sychiatric referral needed?

PATHOLOGICAL ANXIETY

Anxiety is pathological when:

- It is greatly disproportionate to the risks and severity of the stimulus / stressors.
- It continues even when the danger is no longer present.
- Interferes with social, vocational, or physical aspects of daily life.
- Leads to avoidance.

SCREENING FOR GENERALISED ANXIETY DISORDER

Screening for generalised anxiety disorder (GAD) can be done using the Generalised Anxiety 2 item (GAD-2) questionnaire by asking the following 2 questions.

- Over the past 2 weeks, how have you been bothered by the following problems? These are the first 2 questions of GAD-7.

	Not at all	Several days	More than half the days	Nearly everyday
Feeling nervous, anxious or on edge	0	1	2	3
Not being able to stop or control worrying	0	1	2	3

Calculate the sum of the points from the 2 questions in GAD-2. A score of ≥ 3 will proceed in asking the questions in GAD-7 questionnaire (you can also find EPIC flowsheet for the full GAD-7 questionnaire)

	Not at all	Several days	More than half the days	Nearly everyday
Feeling nervous, anxious or on edge	0	1	2	3
Not being able to stop or control worrying	0	1	2	3
Worrying too much about different things	0	1	2	3
Trouble relaxing	0	1	2	3
Being so restless that it's hard to sit still	0	1	2	3
Becoming easily annoyed or irritable	0	1	2	3
Feeling afraid as if something awful might happen	0	1	2	3

Calculate the sum of the points for the 7 questions in GAD-7 to calculate the overall score.

The interpretation of overall score attained for GAD-7 is as follows:

Score	Interpretation
0 – 4 points	No anxiety
5 – 9 points	Mild anxiety
10 – 14 points	Moderate anxiety
15 – 21 points	Severe anxiety

DIAGNOSING THE TYPE OF ANXIETY DISORDER

Type	DSM Diagnostic Criteria
Generalised anxiety disorder	Excessive concern about various situations, happening on more days than not for a minimum of 6 months. The individual struggles against dwelling on the troubling situations. A minimum of 3 of the following symptoms needs to be present (with at least one of the symptoms having been present for the majority of the last 6 months): 1. Restlessness / agitation / keyed up / on the edge 2. Fatigue 3. Trouble concentrating 4. Irritability 5. Muscular tension 6. Sleep disruption (difficulty falling or staying asleep, feeling restless, unsatisfying sleep) The anxiety, worry, or physical symptoms result in significant distress or impairment in daily functioning.
Panic disorder	Repeated panic attacks, with anxiety rapidly escalating within minutes, and must include at least 4 of the following symptoms: 1. Palpitations / rapid heartbeat / pounding heart 2. Sweating 3. Tremors / shaking 4. Breathlessness 5. Choking sensation 6. Chest pain / discomfort 7. Nausea or gastrointestinal distress

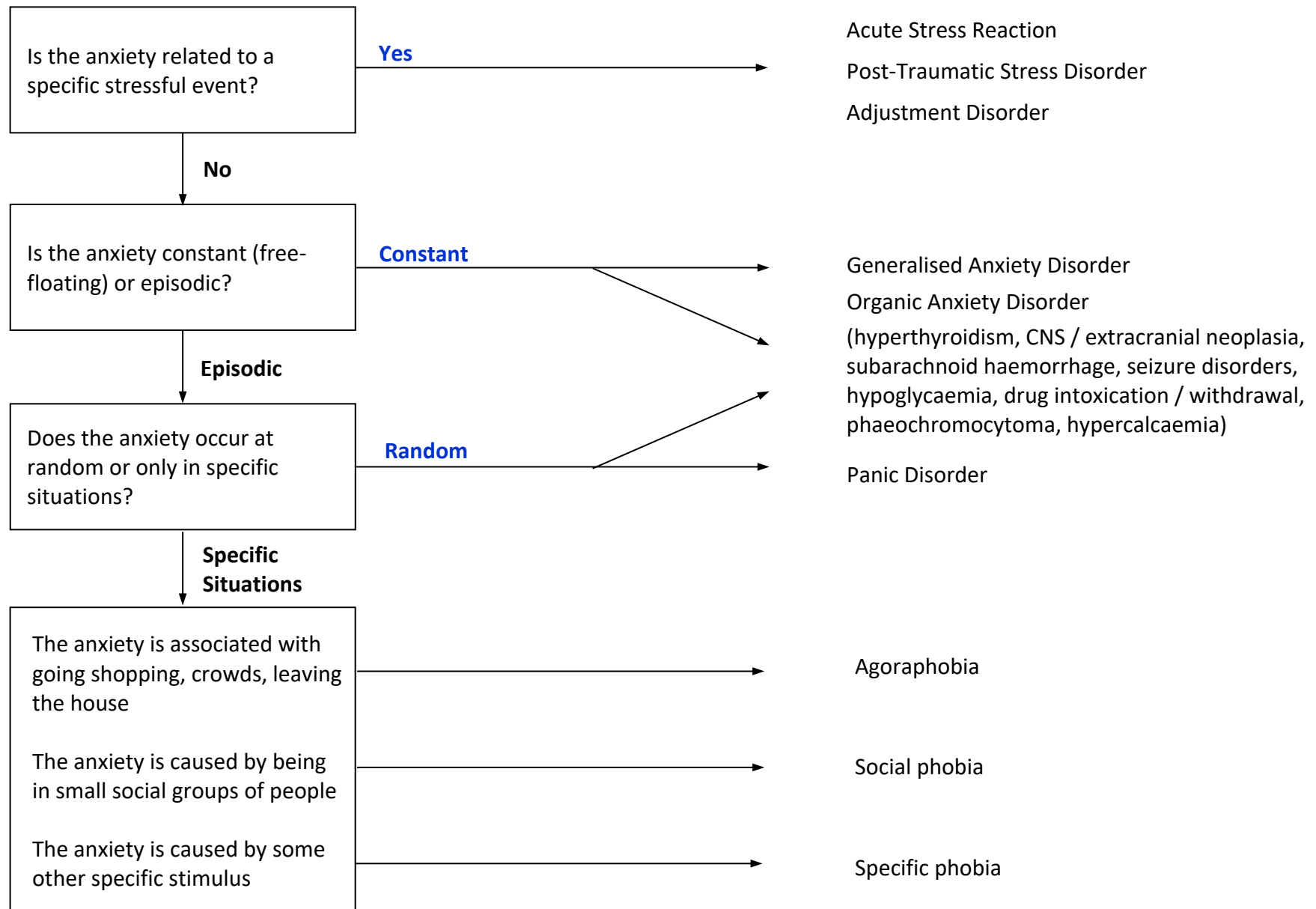
Type	DSM Diagnostic Criteria
	8. Vertigo or sensation of loss of balance, feeling faint, and giddiness 9. Sensations of heat or cold 10. Paraesthesia 11. Derealisation 12. Fear of dying 13. Fear of loss of self-control At least one of these panic attacks must have been followed by at least 1 month or more of 1. Chronic worrying about experiencing additional panic attacks and their possible consequences. 2. Maladaptive behavioural changes related to panic attacks, such as a marked effort to engage in behaviours to avoid panic attacks.
Adjustment disorder	The emergence of emotional or behavioural symptoms in response to specific stressful events, with symptoms occurring within 3 months of the events. Symptoms include one or more of the following: 1. Experiencing severe distress that is out of proportion to the level of stress from the event. 2. A significant decline in important areas of functioning. This stress-related disturbance does not qualify as another mental disorder or a reoccurrence of an existing mental disorder. Once the stressor is removed, the symptoms should not last for more than 6 months.
Acute stress disorder	The symptoms are similar to those of PTSD, but at least 9 symptoms from any category appear right after the trauma and persist for 3 days to 1 month.
Social anxiety disorder /social phobia	Excessive anxiety about being judged by others in various social situations, such as meetings, conversations with unfamiliar people, public speaking, or giving presentations. This fear stems from the worry of being perceived negatively and leads the individual to avoid such situations or endure them with intense anxiety.
Agoraphobia	Intense anxiety about at least two of the following situations: 1. Using public transportation 2. Being in open spaces 3. Being in enclosed spaces 4. Being in a crowd

Type	DSM Diagnostic Criteria
	5. Being alone outside of their home As a result of the fear of not being able to escape from these situations or that assistance is not readily available, the person avoids the above situations. These agoraphobic scenarios typically cause significant distress and anxiety, necessitate the presence of a companion, or are endured while experiencing intense fear.
Post-traumatic stress disorder	The effects of being exposed to actual or potential death, severe injury, or sexual violence, whether through experiencing or witnessing such events, learning that a close friend or family member went through these events, or having repeated exposure to such traumatic events. 1. Experiencing at least one intrusive symptom, such as recurring distressing dreams or flashbacks of a traumatic event. 2. Displaying at least 1 symptom of avoidance, such as trying to avoid distressing thoughts or external stimuli that serve as reminders of the traumatic event. 3. Demonstrating at least 2 negative changes in mood and cognition, such as having negative attitudes about oneself, self-blame, alienation from others, and chronic inability to experience positive emotions. 4. Exhibiting at least 2 symptoms of arousal, such as unprovoked irritability, hypervigilance, heightened startle reactions, and difficulty concentrating. These symptoms must persist for at least one month and may also be accompanied by dissociative symptoms. The onset of the condition may be delayed, with the full diagnostic criteria not being met until at least 6 months after the traumatic event
Specific phobia	An exaggerated fear response to a particular object or situation. The fear is typically triggered immediately by the presence of the feared object or situation, and the person either avoids it or endures it with intense anxiety.

* Anxiety disorders typically last at least 6 months, except that of PTSD and acute stress reactions. The disturbances related to the respective anxiety disorders are not attributed by physiological effects of a substance / drug / medication and not better explained by another mental disorder.

* Please refer to [NUP Obsessive Compulsive Disorder CPG](#) for more information about OCD and its management.

Flowchart for Diagnosis of Anxiety



DIFFERENTIAL DIAGNOSIS FOR ANXIETY SYMPTOMS / MEDICAL CONDITIONS THAT MAY AGGRAVATE ANXIETY SYMPTOMS

The following conditions may aggravate or mimic anxiety symptoms and should be excluded.

Disease System	Examples
Endocrine	Hyperthyroidism, hypoglycaemia, phaeochromocytoma, adrenal insufficiency, hyperadrenocorticism
Cardiovascular	Congestive heart failure, pulmonary embolism, arrhythmia, mitral valve prolapse
Respiratory	Asthma, chronic obstructive lung disease, pneumonia
Metabolic	Diabetes mellitus
Neurologic	Vestibular dysfunction, migraine, neoplasm, temporal lobe epilepsy
Gastrointestinal	Irritable bowel syndrome
Haematologic	Anaemia, Vitamin B ₁₂ deficiency
Drug Misuse	Alcohol withdrawal, benzodiazepine withdrawal, caffeine overuse

PATIENT MANAGEMENT

General Approach is to consider patient needs, preferences, and readiness by discussing and agreeing on:

- Goals of treatment
- Preference between treatment modalities
- Willingness and ability to engage in psychological treatment
- Ability to adhere to regular medication

Non-Pharmacological Therapy

- a) Educate the patient about the nature, aetiology of his / her anxiety symptoms and description of the symptoms of anxiety.
- b) Stress reduction strategies:
 - Deal with negative thoughts
 - Diaphragmatic breathing
 - Progressive muscle relaxation
- c) Encourage exercise.
- d) Suggest mind-body practices, grounding work
- e) Reduce alcohol and caffeine intake. Stop smoking.
- f) Involve family members. Utilise community / social resources.
- g) Supportive counselling and psychotherapy (i.e. cognitive behavioural therapy). Reassure patient that it is frightening and not life threatening.
- h) Monitoring over time and deal with early signs of relapse.
- i) Promote good sleep hygiene.

Pharmacological Therapy

a) Medication for Various Types of Anxiety

Type of Anxiety Disorder	Examples of Non-Pharmacological Therapies	Medication
Generalized Anxiety Disorder*	Cognitive Behavioural Therapy stress management	SSRIs or SNRIs (for generalised anxiety disorders and panic disorders only) Benzodiazepines (for breakthrough anxiety episodes) Sedating anti-histamines (for breakthrough anxiety episodes/ management of agitation / insomnia from SSRIs in the initial phase) Beta-blockers (for palpitations/tremors) 2 nd generation anti-psychotics can be used as augmentation, adding on to anti-depressant for example quetiapine 25mg ON x 1 week → go up to 50mg ON x 3wks and review in 4week's time. If the patient has responded partially to SSRI/ SNRI, 2 nd generation anti-psychotics can be added, especially when higher dosages of SSRI/ SNRI can't be tolerated or already at maximum dose (Will suggest referring to psychiatrist if anti-psychotics are required, unless stepped down from psychiatry)
Panic Disorder[#]	Cognitive Behavioural Therapy (graded exposure)	
Social Phobia	Cognitive Behavioural Therapy (exposure to feared social situation)	
Agoraphobia	Cognitive Behavioural Therapy (graded exposure to feared situation)	
Post-Traumatic Stress Disorder	Multiple modalities e.g. CBT, CPT, EMDR, stress inoculation training, treatment of co-morbid conditions (depression)	
Specific Phobia	Cognitive Behavioural Therapy (exposure to feared situation or object)	Drugs alone are not helpful. Require cognitive behavioural therapy

**For GAD, first line SSRIs are Escitalopram, Paroxetine (not available in NUP formulary) or Sertraline, 2nd line are SNRIs (Venlafaxine)*

[#]For panic disorder, all SSRIs are suitable as first line, 2nd line are SNRIs (Venlafaxine)

b) Drug Classes

For more detailed information on SSRI, SNRIs and TCAs, please refer to [NUP Depression CPG](#).

The following medications below are to use as adjuncts for anxiety or their associated symptoms (i.e. insomnia, palpitations *etc.*) alongside SSRIs:

Benzodiazepines	Common Dose	Maximum Dose	Common ADRs (non-exhaustive list)	Contraindications / Precautions
Alprazolam (0.25 mg tablet)	Short term management of anxiety Initiate at 0.25mg to 0.5mg BD Usual doses: 2 – 6mg/day in 3 – 4 divided doses	10 mg/day in divided doses No renal adjustment required (use with caution) Hepatic adjustment required – (hepatic metabolism)	▪ Sedation ▪ Muscle weakness ▪ Ataxia ▪ Headache ▪ Vertigo ▪ Fatigue ▪ Confusion ▪ Psychomotor impairment ▪ Paradoxical reactions (agitation/insomnia) ▪ Constipation ▪ Increased/decreased appetite ▪ Micturition difficulty	▪ Contraindicated in pregnancy; concomitant use with ketoconazole / itraconazole; narrow-angle glaucoma; pregnancy, severe respiratory insufficiency, myasthenia gravis, sleep apnoea, severe hepatic impairment ▪ Anterograde amnesia (more common with some BZDs e.g. alprazolam) ▪ Avoid in patients with history of alcohol / drug dependence. ▪ Withdrawal rebound anxiety commonly occurs with the shorter acting drugs Librax is not approved for patients below 18 years old ▪ Common interacting medications (especially those with CNS depressing potential e.g. anorex/opioids)
Clonazepam (0.5 mg tablet)	Anxiety Initiate at 0.25 mg BD Usual doses: 1 – 3 mg/day in 1 – 4 divided doses	4 mg/day in divided doses No renal adjustment required (use with caution) Hepatic adjustment required – (hepatic metabolism)		
Lorazepam (0.5 mg / 1 mg tablet)	Anxiety / Insomnia Initiate at 0.5 mg to 1 mg BD	10 mg/day in divided doses No renal adjustment required (use with caution) Hepatic adjustment required – (prolonged elimination half-life)		
Chlordiazepoxide 5mg + clidinium 2.5mg capsules (Librax)	Only for emotional distress caused by irritable bowel syndrome Initiation dose Adult	Adult 2 capsules 4 times a day Geriatric dose 1 capsule 2 times a day		

Beta Blocker	Common Dose	Maximum Dose	Common ADRs (non-exhaustive list)	Contraindications / Precautions
Propranolol (off-label use)	For symptomatic relief of generalised anxiety (palpitations , hand tremors) 10 - 40mg BD to TDS PRN	40mg TDS No renal adjustment required (use with caution) No hepatic adjustment required (use with caution)	▪ Bradycardia ▪ Hypotension ▪ Fatigue ▪ Insomnia ▪ Vivid dreams	▪ Asthma ▪ Bradycardia ▪ hypotension ▪ 1st, 2nd or 3rd degree heart block ▪ May mask the signs and symptoms of hypoglycaemia & hyperthyroidism ▪ Avoid abrupt withdrawal (acute tachycardia/hypertension)

Antihistamines	Common Dose	Maximum Dose	Common ADR	Contraindications / Precautions
Hydroxyzine (10 mg / 25 mg Tablets)	Anxiety (short term use) May initiate at up to 10 - 25 mg/day in divided doses. Dosage adjusted to individual response Hydroxyzine is MOH approved for use in managing anxiety	Max 400mg daily, Max single dose: 100mg for adults 50mg for elderly Renal adjustment required Hepatic adjustment required (may require dosing interval adjustment)	▪ Dizziness ▪ Drowsiness ▪ Headache ▪ Dry mouth ▪ Blurred vision ▪ Constipation ▪ Urinary retention ▪ Hypotension ▪ Thickening of bronchial secretions	▪ Contraindicated during early pregnancy ▪ Caution in urinary retention ▪ Anticholinergic effects are not well tolerated in the elderly ▪ Acute generalized exanthematous pustulosis (rare) ▪ QT prolongation
Diphenhydramine (25mg tab) (off label use)	Short term management of insomnia, rather than for management of anxiety Adult 25 - 50mg given 30 min before bedtime Diphenhydramine is FDA approved for insomnia but not anxiety use			

TREATMENT POINTERS

1. Most Cases of Anxiety Present First to Primary Care Physicians

- Most cases can be managed in primary care
- Take a good history
- Symptoms, severity, duration
- Psychosocial stressors
- Suicide Risk
- Exclude organic illness

2. Some Treatment Pointers

- Start with an SSRI (choice based on side effect profile, drug-drug interactions and/or patient preference / treatment history)
- **Use past responses to medication or psychological treatment to help guide choice of treatment.**
- Early ADRs of SSRI's include agitation & insomnia (faster onset than therapeutic effect on anxiety – Tx with BZDs / hydroxyzine)
- Consider adding B blocker (if severe tremors or palpitations are present)
- If no response, consider a different SSRI before other medications (Adequate trial of SSRI is considered to be 6 weeks at therapeutic dosing)
- Consider doing a baseline ECG for QTc especially when starting escitalopram, venlafaxine, bupropion, amitriptyline, nortriptyline, quetiapine, olanzapine, chlorpromazine, haloperidol.
- Do check sodium after about 3-4 weeks in patients aged > 65 years or those with multiple comorbidities started on SSRI's.
- Listen actively
- Encourage referral to NUP MSW for support/CBT for the mild cases, or NUP Psychologist for the moderate or severe cases.

3. Dosages and Treatment Phases

- Dosages typically similar for adult primary care and psychiatric patients
- Initial drug titration: 4 to 8 weeks
- Monitor closely for emergent suicidal thoughts and behaviour when initiating any antidepressant medication especially those under 25 years of age or with pre-existing suicide risk.
- Revisit safety plan and collaborate with other providers or family / caregivers to ensure support is in place.

4. Maintenance and Remission Phase

- Continuation of treatment for at least 6 to 12 months
- Aim to achieve improvement in symptoms from baseline
- Symptoms are minimal and no longer meet diagnostic criteria
- Restoration of premorbid functioning and improved quality of life

5. Relapse Prevention

- Consider switching to psychological treatment or adding in psychological treatment to medication to reduce risk of relapse once remission is reached.
- Discuss medication discontinuation based on history of relapse, adverse side effect, comorbid mental health conditions, ongoing or anticipated psychosocial stressors, degree of social support and patient's preference.
- Gradually reduce the dose of antidepressant medications to minimise discontinuation symptoms and risk of relapse.
- Risk of relapse is highest in the first several months of stopping treatment, with one-third to half occurring within one year.
- After stopping medication, consider review 1 month after, and 3-6 months after. This could be via video consult.
- Patients should always be educated on how to seek help early should they experience a relapse, and to be aware of their relapse symptoms.

CONSIDERATIONS WHEN USING BENZODIAZEPINES (BZD) IN GENERAL PRACTICE¹

1. The dosage of BZD should be the lowest effective dose necessary to achieve symptomatic relief.
2. Repeat prescriptions for BZD should not be provided without a clinical review.
3. As an adjunct to antidepressant treatment in anxiety disorders, the use of benzodiazepines should be limited to maximally 2 weeks at the lowest effective dose during each consult. The dose should be gradually tapered off. Benzodiazepine use should be closely monitored for adverse effects, abuse, tolerance, dependence and withdrawal symptoms.
4. BZD prescribed for anxiety may be abused by some patients with co-morbid alcohol / substance abuse or dependence and are best avoided where possible in such patients.
5. Do not extend use of benzodiazepines beyond 4 weeks per year at primary care, even when prescribed at the therapeutic dosages.

Specialist referral should be initiated for:

1. Patients who require or have been prescribed BZD beyond a cumulative period of 4 weeks per year.
2. Patients who are already on high dose and or long-term BZD from specialist or hospitals should be referred back to their specialist for review.
3. Patients who are unable to reduce the intake of BZD despite initial commitment to a tapering dose.

Note: Benzodiazepines have no role in the long-term treatment of anxiety disorders. Patients who refuse to be referred to a specialist should be counselled appropriately and documented in the case sheet. If the patient turns aggressive, they should be reported to the police.

ROLE OF HEALTH CARE TEAM MEMBERS

1. Family Physician (Dr)

- Refer patients with psychological problems who fulfilled inclusion criteria to Health and Mind Clinic (HMC) in NUP (those aged ≥ 18 years with depression, anxiety, adjustment, insomnia).
- Manage stable patients discharged from HMC.
- Refer to Psychiatry SOC based on geographical boundaries for those with high risk of suicide or risk of harm to self or others, or patients who are part of the exclusion list of HMC e.g., addictions / legal issues / bipolar disorder / new onset psychosis / new onset OCD.
- Start patients with moderate-severe anxiety disorders on appropriate SSRI's or other appropriate psychotropics (consult senior doctors if unsure).

2. Care Manager in Health and Mind Clinic (HMC CM)

- Gather biodata history and conduct depression, anxiety, insomnia, and suicide risk screening for patients visiting HMC Dr for the first time.
- Offer psychoeducation and basic self-help techniques, along with information and resources related to mental health.
- Conduct virtual consultations to assess the well-being of patients two weeks after initiating treatment with anti-depressants / anti-anxiety medications.
- Conduct virtual consultations to assess well-being of patients after discontinuation of psychotropics to detect early relapse.
- Details to refer to [NUP-WI-CS-COP-030 Management of Patients in Health and Mind Clinic](#).

3. Family Physician in Health and Mind Clinic (HMC Dr)

- Manage new cases and follow-up cases of depression / anxiety / insomnia who are 18 years old and above.
- Follow-up psychiatric stepdown cases.
- Escalate patients to Psychiatry SOC if needed.
- Details to refer to [NUP-WI-CS-COP-030 Management of Patients in Health and Mind Clinic](#).

4. Psychologist

- Conduct psychological screening and assessment during initial assessment to assess the severity of mental health symptoms, complexity of issues and presence / absence of risk tendencies and behaviours.
- Formulate treatment plan for follow-up visits.
- Can work with many types of mental health cases inclusive grief; full list as per [Allied Health - Psych Service](#).

5. Medical Social Worker

- Manage care and counselling inclusive cognitive behavioural therapy if appropriate, for patients with mild anxiety and mild depression.
- Provide sleep hygiene advice for patients with subthreshold insomnia.
- Provide care assessment and arrangement, supportive counselling, crisis intervention, non-medical related financial assistance, and support patients in need of information and referral.

6. Financial Counsellor

- Patients with financial difficulties on medical bills could be referred to Financial Counsellor for assistance.

CONDITIONS FOR REFERRAL TO PSYCHIATRY SOC

Patients should be referred when:

1. Patients 17 years and below who might need psychotropics.
2. If patient has not responded to 3 different antidepressants or very high dose antidepressant monotherapy.
3. Patient requires to take anti-psychotics as an adjunct to managing anxiety disorder (unless patient has been seen by or discussed with psychiatrist prior and is suggested to continue on anti-psychotics)
4. Presence of complicated medical history or special circumstances, e.g. Cushing disease, liver disease *etc.*
5. Presence of comorbid personality disorders, substance dependence.
6. New onset psychotic disorder or bipolar disorder.

RECOMMENDED CARE COMPONENTS FOR ANXIETY

Recommended Care Components	Minimum Frequency*
Generalised anxiety disorder – 7 (GAD-7) score	Every clinical review when appropriate, minimally 6 monthly for patients with generalized anxiety disorder.
Suicide screening	Where clinically indicated – please refer to NUP-WI-CS-COP-032 Management of Suicidal Patients .

** More frequently if clinically indicated*

REFERENCES

1. MOH Clinical Practice Guidelines 2/2008: Prescribing of Benzodiazepines
2. Drug Information Handbook, 17th Edition 2008-2009
3. DSM 5
4. DSM-5-TR
5. Chronic Disease Management Programme – Handbook for Healthcare Professionals. Ministry of Health Singapore 2021